

Projectile Motion at an Angle

Two-dimensional motion — analysis, applications and the weekly assessment, fully worked.

Grade 11 · Egyptian Baccalaureate

27 questions

Step-by-step solutions

g = 10 m/s² unless stated
Mr. Gemy
PHYSICS

The toolkit for this lesson

$$v_{0x} = v_0 \cos \theta$$

constant all the way

$$v_{0y} = v_0 \sin \theta$$

changes by g each second

$$t_{\text{up}} = v_0 \sin \theta / g$$

time to the highest point

$$T = 2 v_0 \sin \theta / g$$

total time of flight

$$H = v_0^2 \sin^2 \theta / 2g$$

maximum height

$$R = v_0^2 \sin 2\theta / g$$

horizontal range

The rule of the lesson. Split the initial velocity into a horizontal part that never changes and a vertical part that behaves exactly like a body thrown straight up. At the highest point $v_y = 0$, but the body is still moving horizontally with $v_0 \cos \theta$.

Analysing the motion of a projectile fired at an angle

DERIVATIONS

- 1 The figure shows a body projected from the ground with an initial velocity v_0 at an angle θ to the horizontal. Write the relation used to calculate each of the following:

a) the horizontal component of the initial velocity (v_{0x}).

b) the vertical component of the initial velocity (v_{0y}).

At position (1): c) the horizontal component v_{x1} · d) the vertical component v_{y1} · e) the magnitude of the resultant velocity v_1 .

f) the horizontal distance X covered from the start up to position (1).

g) the vertical height Y risen from the start up to position (1).

h) the maximum height H reached by the body.

i) the horizontal range R of the body.

j) at position (2): the two components v_{x2} and v_{y2} .

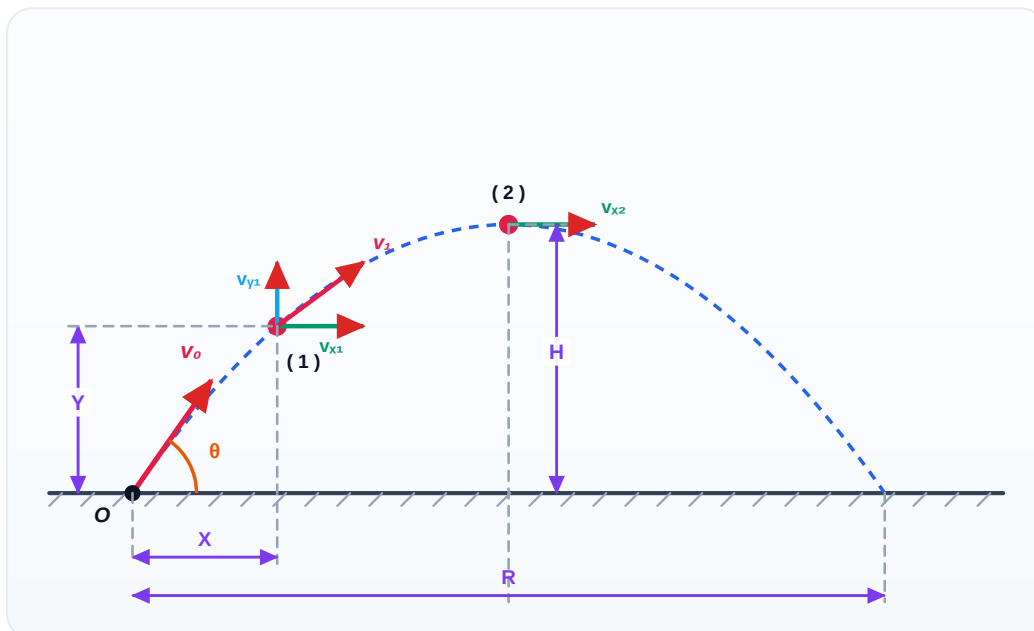


FIGURE 1 · THE COMPLETE PATH OF A PROJECTILE FIRED AT AN ANGLE

HOME ASSIGNMENT 1

The two motions of an angled projectile

MCQ

- 2 The **horizontal** motion of a projectile fired at an angle is characterised by:

A a constant acceleration

B a constant velocity

C an increasing acceleration

D an increasing velocity

3 The horizontal acceleration of a projectile fired at an angle equals:

MCQ

- A zero
- B half the acceleration due to gravity
- C the acceleration due to gravity
- D twice the acceleration due to gravity

4 A small ball is projected from the ground with an initial velocity of 15 m/s at 60° to the horizontal. Neglecting air resistance and taking $g = 10 \text{ m/s}^2$, and knowing that the ball returns to the same level of projection, calculate:

PROBLEM

- a) the time needed to reach the maximum height.
- b) the total time of flight.
- c) the horizontal range of the ball.
- d) the maximum height reached by the ball.

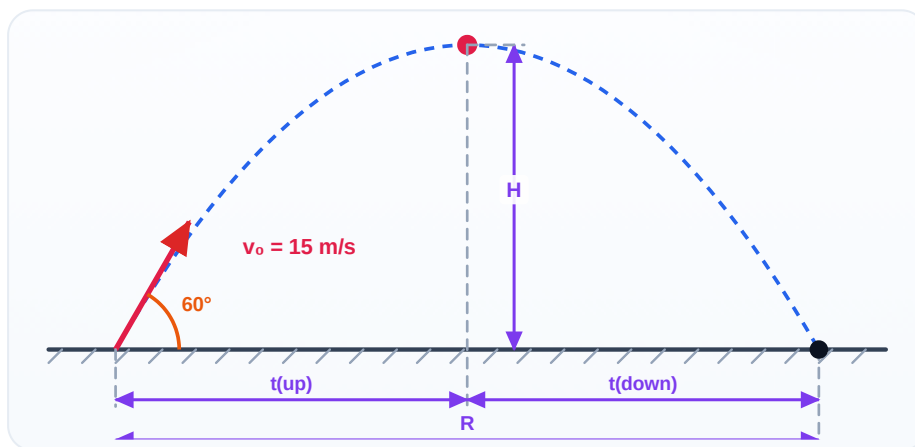


FIGURE 2

CLASSWORK 2

C Applications on projectiles fired at an angle

5 The figure shows a body of mass m projected from the ground with an initial velocity v_0 at 60° to the horizontal, neglecting air resistance. **Write two different ways to increase the total time of flight** of the body.

EXPLAIN

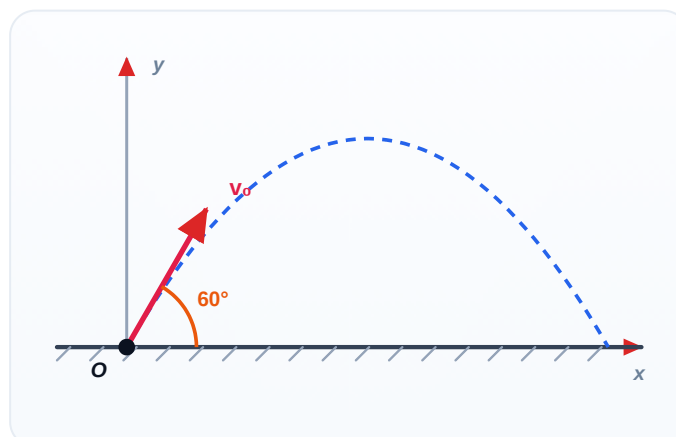


FIGURE 3

6 For the same body of the previous question, write **two different ways** to increase the maximum height reached by the body.

EXPLAIN

7 For the same body, write **two different ways** to increase the horizontal range of the body.

EXPLAIN

8 What is the angle of projection needed for the body to reach its **greatest possible horizontal range**, keeping the initial speed constant? Explain.

EXPLAIN

9 What other angle of projection, used instead of 60° , gives the body the **same horizontal range** shown in the figure, keeping the initial speed constant? Explain.

EXPLAIN

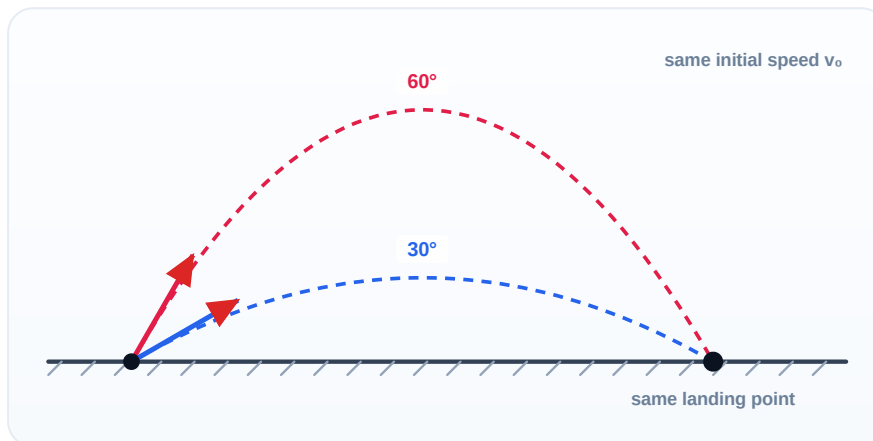


FIGURE 4

The vertical motion and a full calculation

10 The **upward vertical** motion of a projectile fired at an angle is characterised by:

MCQ

- ☐ A a decreasing acceleration
☐ B a decreasing velocity
☐ C an increasing acceleration
☐ D an increasing velocity

11 At the maximum height, the vertical component of the velocity of a projectile fired at an angle equals:

MCQ

- ☐ A zero
☐ B a maximum value
☐ C half the initial vertical component
☐ D the same as the initial vertical component

12 A small ball is projected from the ground with an initial velocity of 20 m/s at 60° to the horizontal. Neglecting air resistance and taking $g = 9.8 \text{ m/s}^2$, and knowing that the ball returns to the same level of projection, calculate:

PROBLEM

- a) the time needed to reach the maximum height.
 b) the total time of flight.
 c) the horizontal range of the ball.
 d) the maximum height reached by the ball.

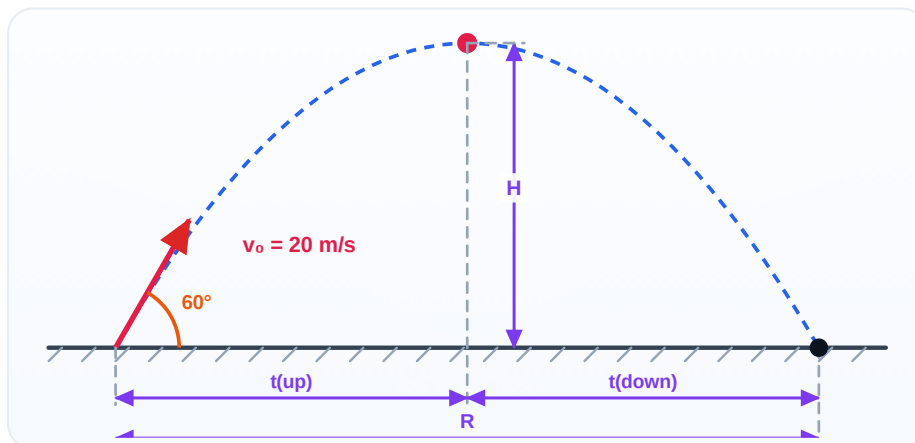


FIGURE 5

Groups A, B and C

Group A

13 At the maximum height, the magnitude of the velocity of a projectile fired at an angle equals:

MCQ

- A zero
- B a maximum value
- C the initial vertical component of the velocity
- D the horizontal component of the initial velocity

14 A body is projected from the ground with a constant initial velocity v_0 at 30° to the horizontal. At which other angle, with the same initial velocity, does it reach the same horizontal range?

MCQ

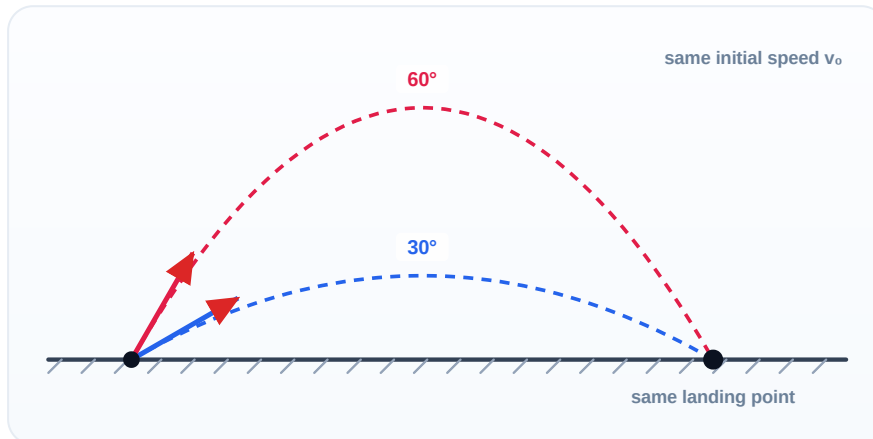


FIGURE 6

- A 30°
- B 45°
- C 60°
- D 75°

15 The figure shows a body of mass m projected from the ground with an initial velocity v_0 at 60° to the horizontal. **Write a way to double the total time of flight** of the body, and explain it.

EXPLAIN

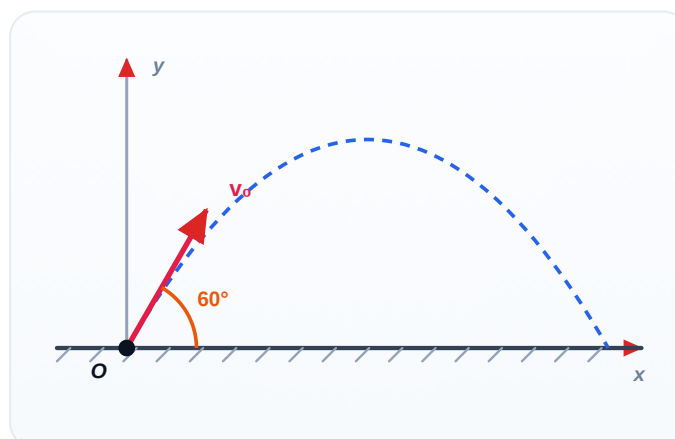


FIGURE 7

- 16 A football is kicked with an initial velocity of 30 m/s at 60° to the **vertical**. Find the horizontal range of the ball.

PROBLEM

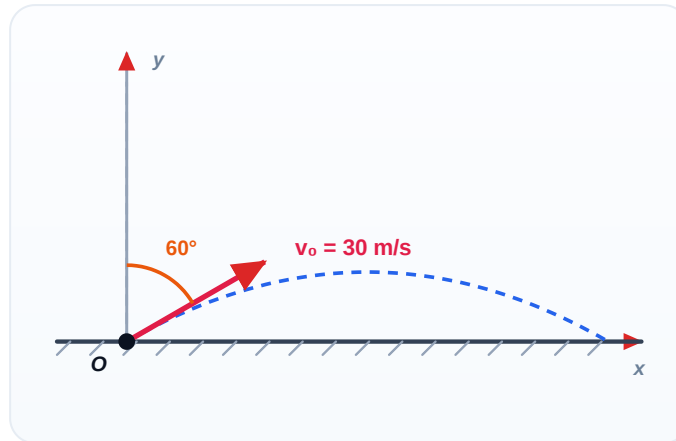


FIGURE 8

- 17 The figure shows a body projected from the top of a building of height 50 m with a velocity of 15 m/s at 30° to the horizontal. **What is the total time of flight** until the body reaches the ground?

PROBLEM

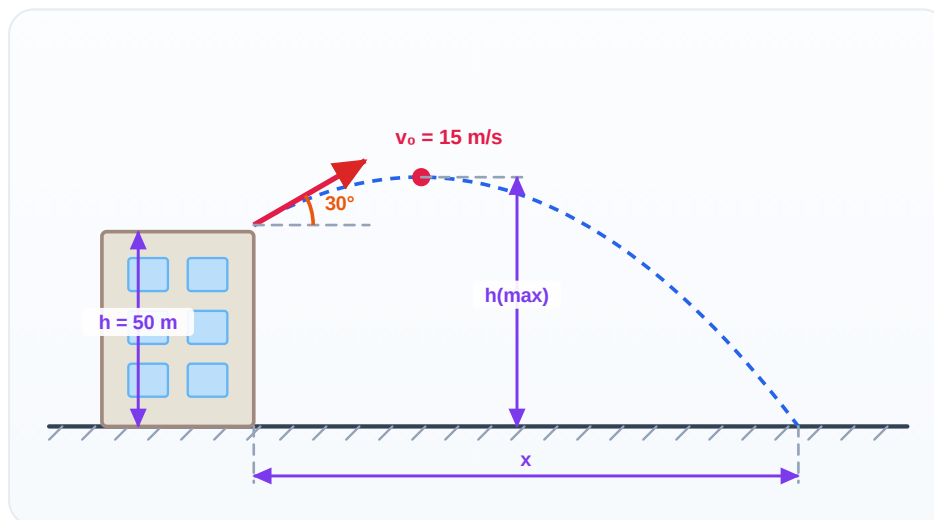


FIGURE 9

Group B

- 18 When a body is projected at an angle to the horizontal, and air resistance is neglected, its **vertical** motion is characterised by:

MCQ

- A a constant velocity and zero acceleration
- B a varying velocity and a constant acceleration
- C an increasing velocity and an increasing acceleration
- D a constant velocity and a constant acceleration

- 19 A body is projected from the ground with an initial velocity v_0 at an angle θ to the horizontal. At which angle θ is the **maximum height equal to the horizontal range**?

MCQ

- A 24° B 45°
C 76° D 90°

- 20 For the body shown in Figure 7, write a way to **double the maximum height** reached by the body, and explain it.

EXPLAIN

- 21 A football is kicked with an initial velocity of 25 m/s at 60° to the horizontal. Find its total time of flight.

PROBLEM

- 22 A body is projected from the top of a building of height 50 m with a velocity of 15 m/s at 30° to the horizontal. **What is the maximum height reached above the ground?**

PROBLEM

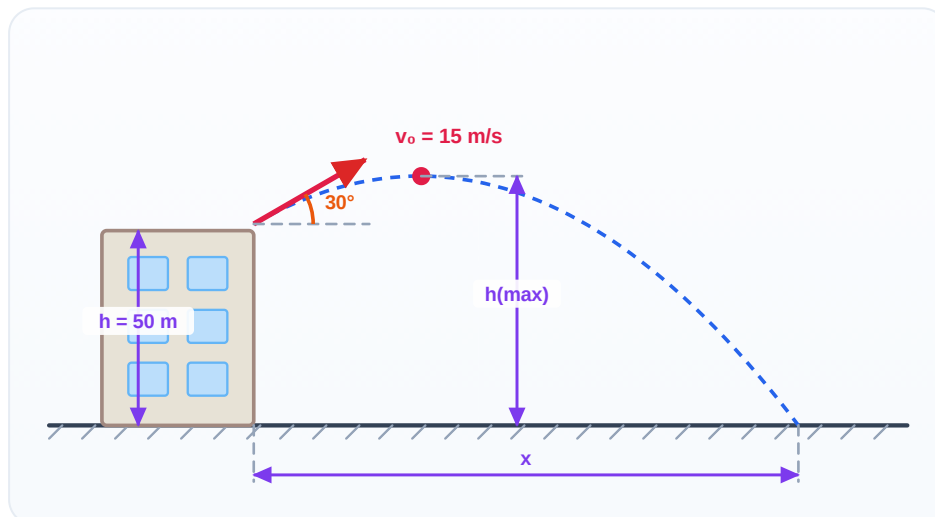


FIGURE 10

Group C

- 23 A body is projected with an initial velocity v_0 at an angle θ to the horizontal, neglecting air resistance. **What happens to the vertical velocity of the projectile while it rises?**

MCQ

- A it increases continuously B it decreases gradually until it becomes zero at the highest point
C it stays constant D it becomes equal to the horizontal velocity

- 24 A body is projected from the ground with an initial velocity v_0 at an angle θ to the horizontal. Which angle of projection gives the projectile the **greatest horizontal range** without changing the speed of projection?

MCQ

A 30°

B 45°

C 60°

D 90°

- 25 For the body shown in Figure 7, write a way to **double the horizontal range** reached by the body, and explain it.

EXPLAIN

- 26 A football is kicked with an initial velocity of 20 m/s at 30° to the **vertical**. Find the maximum height reached by the ball.

PROBLEM

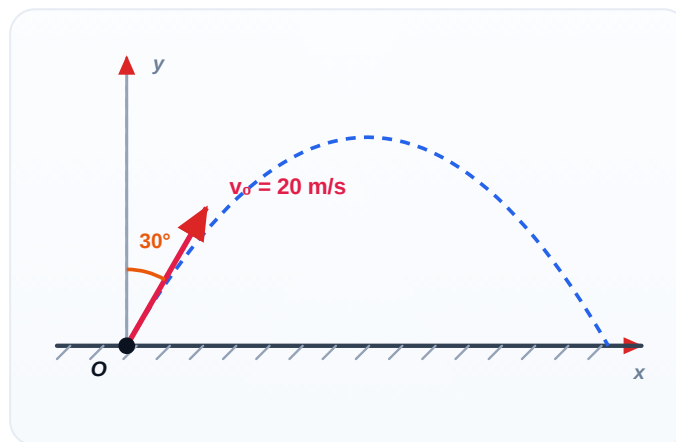


FIGURE 11

- 27 A body is projected from the top of a building of height 50 m with a velocity of 15 m/s at 30° to the horizontal. **What is its horizontal range** measured from the base of the building?

PROBLEM

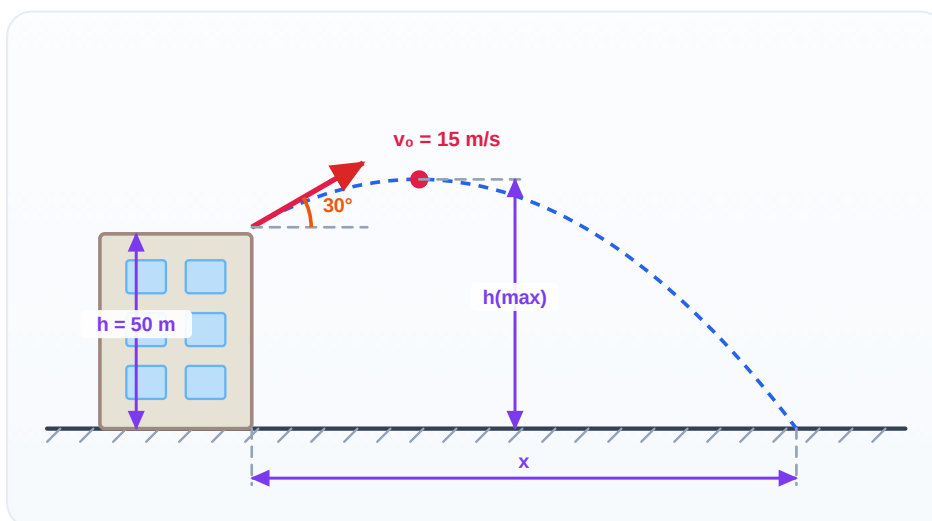


FIGURE 12



ANSWER KEY

Solutions — step by step

The whole lesson comes from two facts: the horizontal velocity never changes, and the vertical motion is a body thrown straight up with an initial speed $v_0 \sin \theta$.

Part A — Classwork 1

1 the relations that describe the whole path

a) $v(\theta_x) = v_0 \cos \theta$ (stays the same all the way)

b) $v(\theta_y) = v_0 \sin \theta$

at position (1) , after a time t_1 :

c) $v(x1) = v_0 \cos \theta$ (no horizontal force)

d) $v(y1) = v_0 \sin \theta - g t_1$

e) $v_1 = \sqrt{v(x1)^2 + v(y1)^2}$

f) $X = (v_0 \cos \theta) t_1$

g) $Y = (v_0 \sin \theta) t_1 - \frac{1}{2} g t_1^2$

h) $H = (v_0 \sin \theta)^2 / (2g) = v_0^2 \sin^2 \theta / (2g)$

i) $R = v_0^2 \sin 2\theta / g$

at position (2) , the highest point :

j) $v(x2) = v_0 \cos \theta$ and $v(y2) = 0$

At the highest point only the vertical component vanishes; the body keeps moving horizontally, which is why its speed there is $v_0 \cos \theta$ and not zero.

Part B — Home assignment 1

2 the horizontal motion

$a(x) = 0 \rightarrow v(x)$ never changes

B) a constant velocity

3 the horizontal acceleration

gravity acts vertically only, so it has no horizontal component

A) zero

4 ball projected at 15 m/s , 60° ($g = 10$)

$$v(\theta x) = 15 \cos 60^\circ = 15 \times 0.5 = 7.5 \text{ m/s}$$

$$v(\theta y) = 15 \sin 60^\circ = 15 \times 0.866 = 12.99 \text{ m/s}$$

a) $t(\text{up}) = v(\theta y) / g = 12.99 / 10 = 1.3 \text{ s}$

b) $T = 2 t(\text{up}) = 2.6 \text{ s}$

c) $R = v(\theta x) \times T = 7.5 \times 2.6 = 19.5 \text{ m}$

d) $H = v(\theta y)^2 / (2g) = 168.75 / 20 = 8.44 \text{ m}$

a) 1.3 s · b) 2.6 s · c) 19.5 m · d) 8.44 m

Part C — Classwork 2

5 doubling down on the time of flight

$$T = 2 v_0 \sin \theta / g \rightarrow T \text{ grows when } (v_0 \sin \theta) \text{ grows}$$

Way 1 : increase the speed of projection v_0 , keeping $\theta = 60^\circ$

Way 2 : increase the angle θ towards 90° , keeping v_0 the same

increase v_0 , or increase θ

6 increasing the maximum height

$$H = v_0^2 \sin^2 \theta / (2g)$$

Way 1 : increase v_0 (H grows with the square of the speed)

Way 2 : increase θ towards 90°

increase v_0 , or increase θ

7 increasing the horizontal range

$$R = v_0^2 \sin 2\theta / g$$

Way 1 : increase v_0

Way 2 : bring the angle closer to 45° (here : decrease θ from 60° towards 45°),
because $\sin 2\theta$ is a maximum at $\theta = 45^\circ$

increase v_0 , or move θ towards 45°

8 the angle of maximum range

$$R = v_0^2 \sin 2\theta / g \text{ is a maximum when } \sin 2\theta = 1$$

$$2\theta = 90^\circ \rightarrow \theta = 45^\circ$$

At 45° the speed is shared equally between the horizontal and the vertical directions, which is the best compromise between staying long in the air and moving fast forward.

$\theta = 45^\circ$

9 the angle that gives the same range

two angles give the same range when they are **complementary** :

$$\sin 2(90^\circ - \theta) = \sin (180^\circ - 2\theta) = \sin 2\theta$$

$$90^\circ - 60^\circ = 30^\circ$$

The 30° throw is flatter and faster, the 60° throw is higher and slower, and the two effects cancel exactly in the range.

$$\theta = 30^\circ$$

Part D — Home assignment 2

10 the upward vertical motion

$v(y) = v_0 \sin \theta - g t \rightarrow$ the vertical velocity decreases steadily
while the acceleration stays constant ($= g$)

B) a decreasing velocity

11 at the maximum height

the body stops rising, so $v(y) = 0$

A) zero

12 ball projected at 20 m/s , 60° ($g = 9.8$)

$$v(0x) = 20 \cos 60^\circ = 10 \text{ m/s}$$

$$v(0y) = 20 \sin 60^\circ = 17.32 \text{ m/s}$$

$$\text{a) } t(\text{up}) = 17.32 / 9.8 = 1.77 \text{ s}$$

$$\text{b) } T = 2 \times 1.77 = 3.54 \text{ s}$$

$$\text{c) } R = 10 \times 3.54 = 35.4 \text{ m}$$

$$\text{d) } H = 17.32^2 / (2 \times 9.8) = 300 / 19.6 = 15.3 \text{ m}$$

a) 1.77 s · b) 3.54 s · c) 35.4 m · d) 15.3 m

Part E — Weekly assessment 3 · Group A

13 the speed at the top

at the highest point $v(y) = 0$, so

$$v = v(x) = v_0 \cos \theta$$

D) the horizontal component of the initial velocity

14 the complementary angle

$$90^\circ - 30^\circ = 60^\circ \text{ gives the same range}$$

C) 60°

15 doubling the time of flight

$T = 2 v_0 \sin \theta / g$, with θ fixed at 60° : T is proportional to v_0
so the speed of projection must be doubled ($v_0 \rightarrow 2 v_0$)

Raising the angle cannot do it here: T would need $\sin \theta$ to double, and $2 \sin 60^\circ$ is greater than 1, which is impossible for any angle.

double the initial speed of projection

16 an angle measured from the vertical

the angle with the horizontal = $90^\circ - 60^\circ = 30^\circ$

$$R = v_0^2 \sin 2\theta / g = (30)^2 \times \sin 60^\circ / 10$$

$$R = 900 \times 0.866 / 10 = 77.9 \text{ m}$$

Always convert an angle given with the vertical into an angle with the horizontal before using the projectile relations.

R = 77.9 m

17 projected from the top of a building

$$v(0x) = 15 \cos 30^\circ = 12.99 \text{ m/s}$$

$$v(0y) = 15 \sin 30^\circ = 7.5 \text{ m/s}$$

taking upwards as positive, the final vertical displacement is -50 m :

$$-50 = 7.5 t - 5 t^2$$

$$5 t^2 - 7.5 t - 50 = 0 \quad \div 5 \quad \rightarrow \quad t^2 - 1.5 t - 10 = 0$$

$$t = [1.5 + \sqrt{(2.25 + 40)}] / 2 = (1.5 + 6.5) / 2 = 4 \text{ s}$$

$T = 2 v_0 \sin \theta / g$ cannot be used here, because the body does not return to the level it started from.

T = 4 s

Group B

18 the vertical motion

$v(y)$ changes every second, while $a = g$ is constant in size and direction

B) a varying velocity and a constant acceleration

19 when the height equals the range

$$H = R$$

$$v_0^2 \sin^2 \theta / (2g) = v_0^2 \sin 2\theta / g = 2 v_0^2 \sin \theta \cos \theta / g$$

$$\sin^2 \theta = 4 \sin \theta \cos \theta \quad \rightarrow \quad \tan \theta = 4$$

$$\theta = 76^\circ$$

C) 76°

20 doubling the maximum height

$H = v_0^2 \sin^2 \theta / (2g)$, with θ fixed : H is proportional to v_0^2
to double H : v_0^2 must double $\rightarrow v_0 \times \sqrt{2}$ (about $1.41 v_0$)

Doubling the speed itself would make the height **four** times as large, not twice.

multiply the speed of projection by $\sqrt{2}$

21 time of flight of the football

$v(0y) = 25 \sin 60^\circ = 21.65 \text{ m/s}$

$T = 2 v(0y) / g = 2 \times 21.65 / 10 = 4.33 \text{ s}$

T = 4.33 s

22 the maximum height above the ground

$v(0y) = 15 \sin 30^\circ = 7.5 \text{ m/s}$

height risen above the roof : $H = v(0y)^2 / (2g) = 56.25 / 20 = 2.81 \text{ m}$

height above the ground = $50 + 2.81 = 52.81 \text{ m}$

The relation for H always measures the rise **above the point of projection**, so the height of the building must be added afterwards.

h(max) = 52.81 m above the ground

Group C

23 the vertical velocity while rising

$v(y) = v_0 \sin \theta - g t$ keeps falling until it reaches zero at the top

B) it decreases gradually until it becomes zero at the highest point

24 the angle of greatest range

$R = v_0^2 \sin 2\theta / g$ is a maximum when $2\theta = 90^\circ$

B) 45°

25 doubling the horizontal range

$R = v_0^2 \sin 2\theta / g$, with θ fixed at 60° : R is proportional to v_0^2
to double R : v_0^2 must double $\rightarrow v_0 \times \sqrt{2}$

Changing the angle alone cannot double it: the largest possible value of $\sin 2\theta$ is 1, while $\sin 120^\circ$ is already 0.87, so the angle can add at most about 15 %.

multiply the speed of projection by $\sqrt{2}$

26 kicked at 30° to the vertical

the angle with the horizontal = $90^\circ - 30^\circ = 60^\circ$

$$v(0y) = 20 \sin 60^\circ = 17.32 \text{ m/s}$$

$$H = v(0y)^2 / (2g) = 300 / 20 = 15 \text{ m}$$

$$H = 15 \text{ m}$$

27 the range from the top of the building

$$v(0x) = 15 \cos 30^\circ = 12.99 \text{ m/s}$$

the total time of flight was found in question 17 : $t = 4 \text{ s}$

$$x = v(0x) \times t = 12.99 \times 4 = 51.96 \text{ m}$$

$$x \approx 52 \text{ m}$$